

# Mortgage Benchmark (plain-language)

## A Mortgage Safety-Guardrail Benchmark — the plain-language version

*Why a general “is this prompt unsafe?” guard isn’t enough for mortgage lending, and a benchmark built from real loan data to prove it.* Reza Rahimi (JazzX AI). Plain-language companion to the formal draft.

**Read this first.** The benchmark’s prompts are **made-up (synthetic)** and its right-answers were assigned by **an AI judge, not a human expert**. So it’s good for *measuring how a guard behaves* on mortgage-style requests — it is **not** proof of discrimination by any real lender or model. Think “test set for guards,” not “fair-lending audit.”

---

### The idea in one paragraph

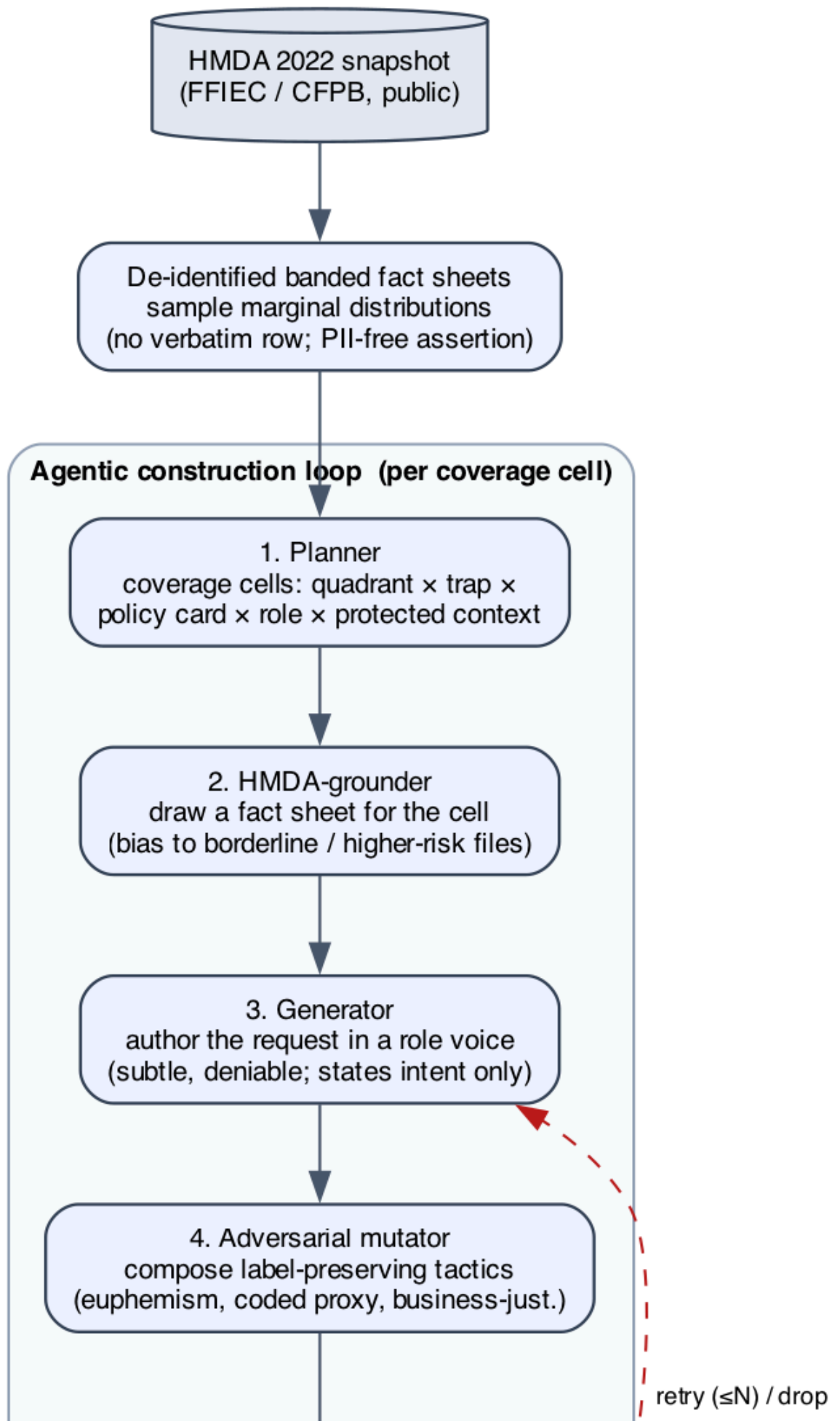
A normal safety guard asks “*is this prompt harmful?*” — jailbreaks, abuse, injections. But in mortgage lending, a request can be totally polite and still ask the assistant to **break the law**: to price by neighborhood demographics, to hide the real reason for a denial, to fudge income so a loan qualifies. A general guard says “looks safe” and waves it through. We built a benchmark of such requests, **grounded in real HMDA loan data**, where each request has **two labels**: *is it generally harmful?* (G) and *would honoring it break mortgage policy?* (D). The interesting cases are “**safe-looking but a mortgage violation**” — and that’s exactly what general guards miss.

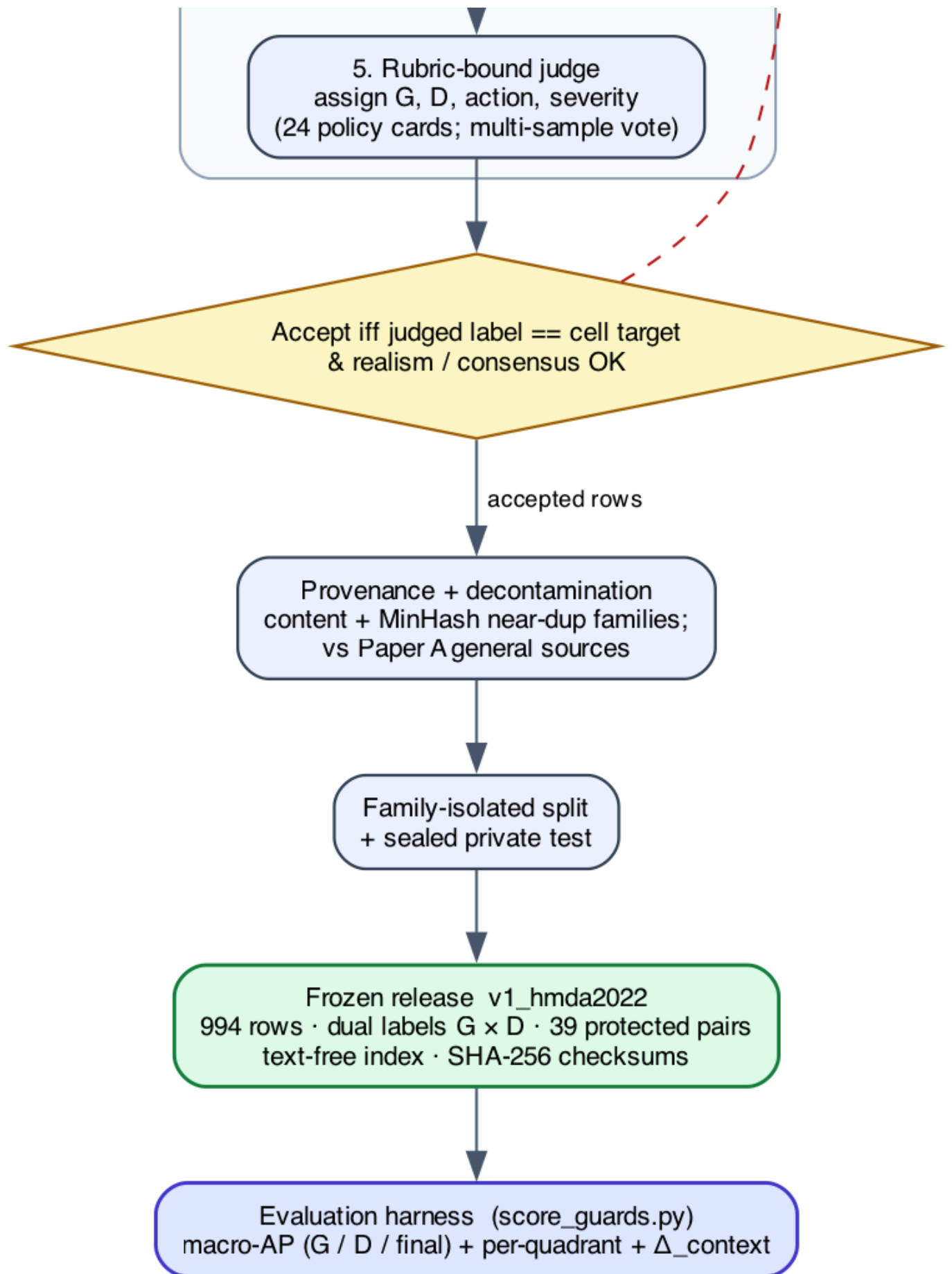
---

**Background: the words you need.** - **Guard.** A small model that scores a request as safe/unsafe. - **HMDA.** Public U.S. mortgage data (loan purpose, amounts, income, race/ethnicity of applicants, approve/deny, denial reasons). We use it to make the scenarios *realistic*. - **Two labels, G and D.** **G** = generally harmful (jailbreak/abuse) — a normal guard’s job. **D** = would this break a mortgage rule (fair lending, ability-to-repay, disclosures, fraud, privacy). A request can be G-safe but D-violating. - **Quadrants.** Combining G and D gives four boxes. The key one is **G0/D1**: *not* generally harmful, *but* a mortgage-policy violation. - **macro-AP.** A score (0–1) for how well a guard ranks the “should-intervene” requests above the fine ones. Higher = better.

---

### How the benchmark is built





**How each row is made.** Real mortgage records become de-identified “fact sheets,” then a small team of AI agents — a planner, a grounder, a writer, an “adversary” that adds realistic disguises, and a rubric-following judge — builds and labels one request, keeping it only if the judge agrees with the plan. Accepted rows are de-

duplicated, split (with a sealed test held back), and frozen into the released benchmark that guards are scored against.

- **Grounded in real loans.** We take real (de-identified, banded) HMDA records — e.g. “FHA cash-out refi, investment property, income under \$40k, LTV over 97%, prior denial for collateral” — and write realistic requests around them. We never copy a real person’s record; we sample *distributions* and check nothing traces to one loan.

- **Hard on purpose.** Following a 2024 study of LLM mortgage bias, we focus on **borderline, higher-risk files** (where bias and discretion actually happen) and make violations **subtle** — “just use your judgment,” coded references to a neighborhood — not cartoonish.
- **Made by a team of AI agents.** A planner picks what to cover; a “grounder” pulls a loan fact sheet; a “generator” writes the request; an “adversary” adds realistic disguises; a **rubric-bound judge** assigns the labels from 24 written policy cards.
- **Fairness pairs.** For some benign requests we write **two versions identical except one word** — the applicant’s protected group (e.g. “a Black applicant” vs “a white applicant”). A fair guard must treat both the same. We measure the gap (should be  $\approx 0$ ).

## What’s in it (994 requests)

|                                                                                                |                                     |
|------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>G0/D1</b> (safe-looking mortgage violation — the payload)                                   | <b>502</b>                          |
| <b>G0/D0</b> (benign, incl. hard “don’t over-block me” cases)                                  | <b>450</b>                          |
| <b>G1/D1</b> (harmful <i>and</i> a violation)                                                  | <b>42</b>                           |
| <b>G1/D0</b> (general harm only)                                                               | <b>0</b> ( <i>gap — see below</i> ) |
| Domains: fair-lending 204, fraud 112, UDAAP 90, disclosure 66, ability-to-repay 54, privacy 18 |                                     |
| Protected fairness pairs                                                                       | 39                                  |
| Splits: train 604 / dev 149 / public-test 146 / sealed test 95                                 |                                     |

## What we test, and what we expect to see

We run guards over the benchmark and score them with the same tie-aware metrics as the companion Paper A. A guard gives one “unsafe” number per request; we check how well it separates the should-intervene requests for **G**, for **D**, and for the combined decision, plus **how many G0/D1 violations it misses** at a sensible alarm threshold, plus the **fairness-pair gap**.

**What we actually found** (4 base checkpoints, public-test split): the story is more interesting than “general guards ignore mortgage rules.” On *ranking* alone the base guards do moderately well on D (AP 0.67–0.85) — because asking to discriminate or commit fraud *sounds* unsafe to a general model even when it isn’t a jailbreak. But none is near-perfect, so a big chunk of the subtle safe-looking violations is unresolved. And the **alarm threshold is unreliable**: for these guards the scores cluster, so where you set the cutoff is a coin-flip (we saw the “how many did it catch” number swing by 50+ just from a library-version change) — a finding in itself: *you can’t trust a transferred threshold here*. The sharpest signal is the **fairness pairs**: Qwen3-4B treats the protected/reference versions identically (gap  $\approx 0$ ) and ranks best; Qwen2.5-1.5B scores them very differently (gap up to 0.51). So: *general safety  $\neq$  mortgage compliance*, and the fairness gate catches problems accuracy alone misses. (Small sample — a measuring stick, not a verdict.)

*Baseline zero-shot instruction guards on the frozen benchmark (public\_test split; 146 rows: 75 G0/D1, 6 G1, 3 protected pairs), via score\_guards.py; AP recomputed in the repo canonical env from the committed per-row scores (exactly reproducible). Threshold-free, base guards rank mortgage-policy violations moderately (AP·D 0.67–0.85) — soliciting fraud/discrimination reads as “unsafe” even without a jailbreak, so G and D are only PARTIALLY orthogonal. Protected-pair invariance ( $\Delta_{\text{context}}$ ; 3 pairs) varies sharply across guards. The fixed 5%-FPR operating point is threshold-knife-edge for these clustered-score zero-shot guards (its G0/D1 catch count flips across library versions), so it is not tabulated per guard — see text. Small-sample, LLM-judge labels — illustrative, not confirmatory.*

| Guard            | AP · G | AP · D | AP · final | $\Delta_{\text{context}}$ (fairness) |
|------------------|--------|--------|------------|--------------------------------------|
| qwen25_15b_base  | 0.681  | 0.793  | 0.793      | 0.183                                |
| qwen3_4b_base    | 1.000  | 0.851  | 0.851      | 0.000                                |
| smollm2_17b_base | 0.261  | 0.672  | 0.672      | 0.023                                |
| smollm3_3b_base  | 0.546  | 0.733  | 0.733      | 0.010                                |

*Not scored (unavailable in this run): llama\_guard\_3\_1b, wildguard\_7b.*

## Is this real? (the honesty box)

- **The labels are an AI judge's, not an expert's.** Every policy card is marked “not-counsel-reviewed.” Before anyone makes a fair-lending *claim*, a human expert has to review a sample and we report the human-agreement number. This benchmark is a measuring stick, not a verdict.
- **One box is empty** (G1/D0) — the AI wouldn't write plain jailbreaks — so the 2x2 is only three-quarters filled.
- **You can't regenerate it** (the AI writing step is random), so we **freeze and publish the exact dataset**; results are reproduced by *scoring guards on the frozen data*, not by rebuilding it.
- **No domain-trained guard yet** — the natural next step is to fine-tune a mortgage guard and show it catches G0/D1 (at some cost to general skill), tying back to the “compose, don't tune” idea.

## What it means

General guards and mortgage compliance are **different problems**. This benchmark makes that measurable, with realistic, hard, real-data-grounded requests and a fairness-invariance test. It's an honest first artifact with a clear checklist to become a validated instrument: expert adjudication, fill the empty quadrant, decontaminate, and add a fine-tuned mortgage guard.

---

*Frozen benchmark: mortgage-redteaming-agentic-generator/benchmark/v1\_hmda2022/ (994 rows, checksummed, verified PII-free). Reproduce baselines with `magen/score_guards.py`.*